

EEE4114F Class Test 2

May 2024

Duration: 1 hour

Total: 40 marks

Empl ID: _____

Instructions:

- This is a closed-book exam.
- There are two parts to this exam.
- **Part A** has 2 questions totalling 20 marks. You must answer all of them.
- **Part B** has 2 questions totalling 20 marks. You must answer all of them.
- Both parts can be answered in the space provided on the exam script, and you can write on the back of the printed pages if required. Answer books will be available should you need additional writing space.
- You must use a pen to complete the exam — pencil will only be marked for drawings. Values indicated on drawings must be written in pen. Please show all of your working clearly — method and approach matter.
- A table of standard Fourier transform and z-transform pairs is provided separately
- You have 1 hour.

Question	Marks	Grade
1	10	
2	10	
3	10	
4	10	
Total:	40	

Part A: Digital Signal Processing

Question 1 [10 marks]

A system has impulse response $h[n] = (1/2)^n u[n]$. Determine the input $x[n]$ to the system if the output is given by

$$y[n] = \frac{1}{3}u[n] + \frac{2}{3}\left(-\frac{1}{2}\right)^n u[n].$$

Solution: The z-transform of the output is

$$Y(z) = 1/3 \frac{1}{1 - z^{-1}} + 2/3 \frac{1}{1 + 1/2 z^{-1}}$$

with ROC $|z| > 1$.

The system function is

$$H(z) = \frac{1}{1 - 1/2 z^{-1}}$$

with ROC $|z| > 1/2$.

The z-transform of the input is

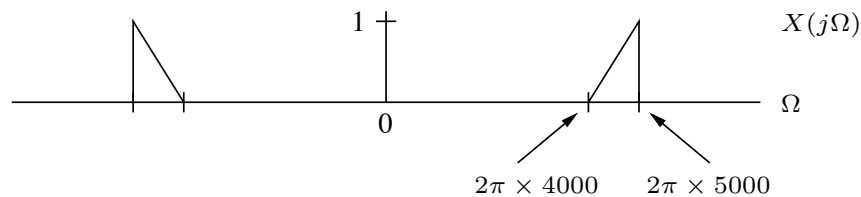
$$X(z) = \frac{Y(z)}{H(z)} = 1/3 \frac{1 - 1/2 z^{-1}}{1 - z^{-1}} + 2/3 \frac{1 - 1/2 z^{-1}}{1 + 1/2 z^{-1}}.$$

This has poles at $z = 1$ and $z = -1/2$, so the three possible ROCs are $|z| < 1/2$, $1/2 < |z| < 1$, and $|z| > 1$. We require that ROC_Y contain $\text{ROC}_H \cap \text{ROC}_X$, which implies that ROC_X is $|z| > 1$. The inverse is the input:

$$h[n] = 1/3 u[n] - 1/6 u[n-1] + 2/3 (-1/2)^n u[n] - 1/3 (-1/2)^{n-1} u[n-1].$$

Question 2 [10 marks]

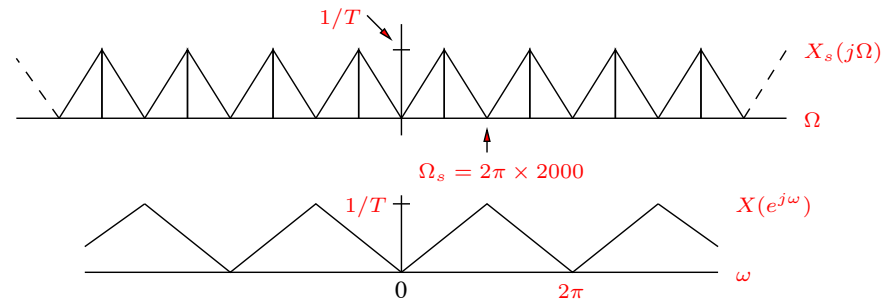
A continuous-time signal $x_a(t)$ has the following Fourier transform:



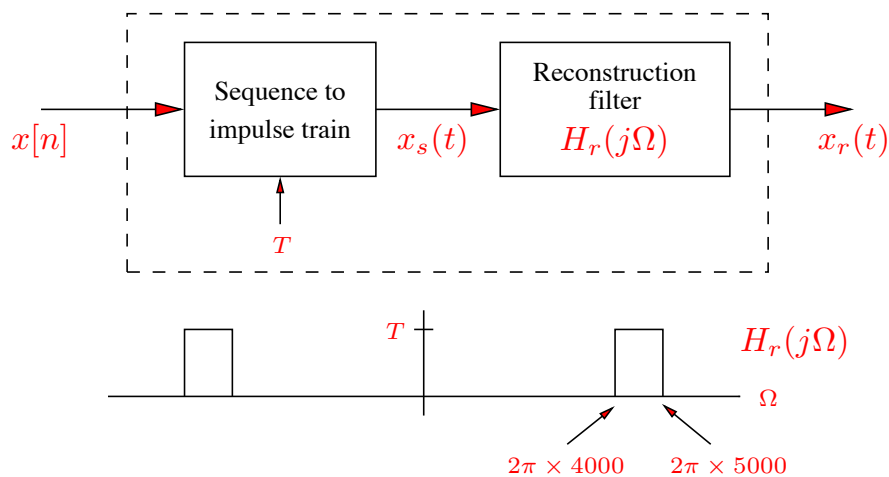
1. What is the highest frequency present in the signal?
2. What is the lowest sampling frequency that can be applied to the signal such that no aliasing occurs and it can still be reconstructed? Draw the DTFT of $x[n]$ in this case.
3. Draw the block diagram of a reconstruction system that will recover $x_a(t)$ from the sampled signal $x[n]$ if the sampling rate is set to its lowest possible value. Sketch the frequency response of the reconstruction filter to use in this case.

Solution:

1. The highest frequency present in the signal is $\Omega = 2\pi \times 5000$ rad/s.
2. If we digitise with a sampling frequency of $\Omega_s = 2\pi \times 2000$ rad/s, or in other words with $T = 1/2000$ s, then there is no spectral overlapping or aliasing and the signal can be recovered. See plots below:



3. The original signal can be reconstructed from the sampled signal $x[n]$ as follows:



Part B: Machine Learning

Question 3 [10 marks]

Suppose you train three different models: A, B, and C, as part of a task to predict the output power of a wind turbine based on wind speed. A plot of each model's predictions compared to the training data collected from a wind turbine is shown in Figure 1.

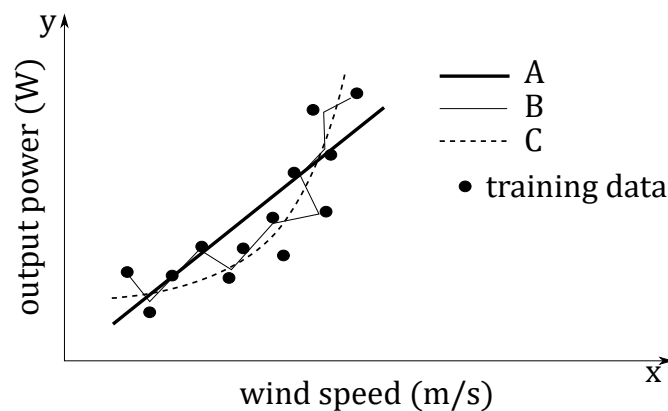


Figure 1: Plot of model predictions compared to training data

1. Is there sufficient information in Figure 1 to determine if model C would work well if deployed in a real-world wind farm? (2)

Solution: No, comparing results to training data alone is not indicative to performance on unseen data. There should at least be some data not used to train the model to evaluate it.

2. Figure 2 shows a plot of model capacity vs error. Which points on this plot (points indicated as 1, 2, or 3 on the x-axis) would you expect to correspond to each model (A, B, and C)? Explain your decisions and discuss the performance of each model with respect to bias and variance. (6)

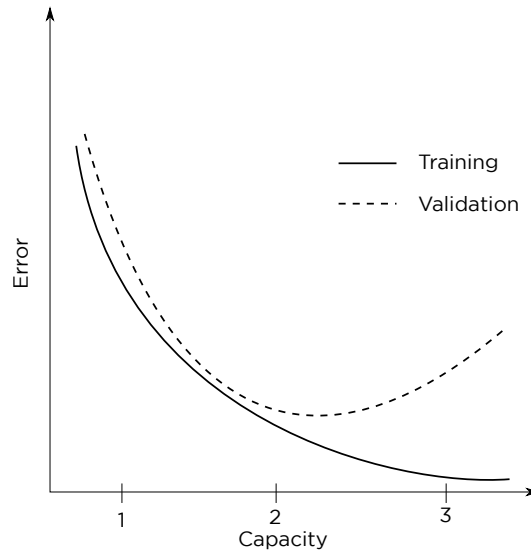


Figure 2: Model capacity vs. error

Solution:

- 1 Model A, the model performs poorly on the training data which would imply that it does not have the capacity to learn the relationship. This model would have high bias and low variance, as it is unable to estimate the training data with reasonable effectiveness, but it is unlikely to change significantly with different data from the same distribution.
- 2 Model C, the model performs well on the training data which is an indication of low bias and appears to have an output that would generalize better to unseen data which is an indication of low variance.
- 3 Model B, the model performs very well on the training data which is an indication of low bias, however it appears to have an output that is less likely to generalize to unseen data compared to model C, which is an indication of high variance.

3. What could be done to improve model B without changing the number of model parameters and without adding additional training data? (2)

Solution: Regularization could be used to constrain the model parameters without changing how many there are during training. Model B's output indicates that it has high flexibility/capacity which results in low bias, but will lead to high variance. Regularization would attempt to constrain the model parameters to small values which would reduce the effect of certain parameters on the response. The result is effectively a simpler model that should perform better on unseen data

Question 4 [10 marks]

Suppose you need to develop a system that estimates the number of fruits a tree will produce based on the concentration of Potassium in the soil collected by a sensor buried in the ground. You collect a data set, $\mathcal{D}$, consisting of input-output pairs (x_i, y_i) . x_i is the concentration of Potassium measured in the soil for a single tree. y_i is the target variable, which, in this case, is the number of fruits collected from that tree.

$$\mathcal{D} = \{(1.1, 10), (2.0, 20), (1.4, 15), (2.8, 30)\}$$

You then decide to fit a regression model of the form, $h_\theta(x) = \theta_0 + \theta_1 x^2$, to it using gradient descent optimization methods. The objective function, J , you choose to optimize is given by the mean square error of the form:

$$J = \frac{1}{2N} \sum_{i=1}^N (h_\theta(x_i) - y_i)^2$$

Where N is the number of examples in a single training iteration.

The model's parameters are randomly initialised such that

$$\theta_0 = 0, \theta_1 = 1$$

Suppose you decide to divide the data set into two, equally sized mini-batches, keeping the order as it is such that

$$\mathcal{D}_1 = \{(1.1, 10), (2.0, 20)\}, \mathcal{D}_2 = \{(1.4, 15), (2.8, 30)\}$$

A single training iteration uses one mini-batch and the model parameters are updated after **each** iteration. The update rule is given by:

$$\theta_i := \theta_i - \eta \frac{\partial J}{\partial \theta_i}$$

Where $\eta = 0.1$. Answer the following questions:

1. Write down the expressions for the gradients of the objective function with respect to the model's parameters $\frac{\partial J}{\partial \theta_0}$ and $\frac{\partial J}{\partial \theta_1}$. (2)

Solution:

$$\begin{aligned} \frac{\partial J}{\partial \theta_0} &= \frac{1}{2} \left(\sum_{i=1}^2 ((\theta_0 + \theta_1 x_i^2) - y_i) \right) \\ \frac{\partial J}{\partial \theta_1} &= \frac{1}{2} \left(\sum_{i=1}^2 ((\theta_0 + \theta_1 x_i^2) - y_i) x_i^2 \right) \end{aligned}$$

2. Calculate the gradients of the parameters after one training iteration using the first mini-batch, $\mathcal{D}_1$. (2)

Solution:

$$\begin{aligned} h_\theta(1.1) &= 0 + 1(1.1^2) = 1.21 \\ h_\theta(2.0) &= 0 + 1(2.0^2) = 4 \\ \frac{\partial J}{\partial \theta_0} &= \frac{1}{2} ((1.21 - 10) + (4 - 20)) = -12.4 \\ \frac{\partial J}{\partial \theta_1} &= \frac{1}{2} ((1.21 - 10) + (1.1^2) + (4 - 20)(2^2)) = -37.32 \end{aligned}$$

3. Calculate what the model would be parameters after the **second** training iteration. (4)

Solution: First, we need to know the parameters after the first update.

$$\begin{aligned}\theta_0 &:= \theta_0 - \eta \frac{\partial J}{\partial \theta_0} \\ &:= 0 - 0.1(-12.4) \\ &:= 1.24\end{aligned}$$

$$\begin{aligned}\theta_1 &:= \theta_1 - \eta \frac{\partial J}{\partial \theta_1} \\ &:= 1 - 0.1(-37.32) \\ &:= 4.73\end{aligned}$$

Then repeat the calculation of gradients, but using the new parameters and new data $h_\theta(x_i) = 1.24 + 4.73(x_i)^2$

$$h_\theta(1.4) = 1.24 + 4.73(1.4^2) = 10.54$$

$$h_\theta(2.8) = 1.24 + 4.73(2.8^2) = 38.35$$

$$\frac{\partial J}{\partial \theta_0} = \frac{1}{2}((10.54 - 15) + (38.35 - 30)) = 1.95$$

$$\frac{\partial J}{\partial \theta_1} = \frac{1}{2}((10.54 - 15)(1.4^2) + (38.35 - 30)(2.8^2)) = 28.36$$

Then the new parameters are

$$\begin{aligned}\theta_0 &:= \theta_0 - \eta \frac{\partial J}{\partial \theta_0} \\ &:= 1.24 - 0.1(1.95) \\ &:= 1.08\end{aligned}$$

$$\begin{aligned}\theta_1 &:= \theta_1 - \eta \frac{\partial J}{\partial \theta_1} \\ &:= 4.73 - 0.1(28.36) \\ &:= 1.89\end{aligned}$$

4. Is the choice of hypothesis function appropriate for the data? Would this result in high or low bias? (2)

Solution: Not appropriate. The targets, y , appear to be linear with respect to x . This would result in high bias, as the model would be unable to replicate the 'true' function in its current form.